

PAMPAr-o1 v9: A Brain-Inspired Territorial Architecture for Language Modeling with Explicit Rule-Based Routing

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Abstract

We introduce PAMPAr-o1 v9, a novel language model architecture inspired by the functional organization of the human brain. Unlike conventional transformer models that rely solely on learned attention mechanisms, PAMPAr-o1 employs a hybrid routing system combining explicit rule-based token classification (LLAVES, 70%) with learned attention (30%). The architecture organizes computation into four specialized *territories*—Expressive, Contextual, Formal, and Structural—coordinated by a central *Thalamus* module that routes tokens based on their linguistic properties. Territories communicate through six bidirectional *frontiers* with learned gating mechanisms. Preliminary experiments on WikiText-103 demonstrate that PAMPAr-o1 v9 achieves a perplexity of approximately 45 with only 14M parameters, outperforming LSTM baselines (69.1 PPL, 24M params) and approaching Transformer-XL Small performance (54.5 PPL, 24M params) with 42% fewer parameters. We release all code under AGPL-3.0-or-later license to encourage further research into biologically-inspired language architectures.

Keywords: language modeling, brain-inspired architecture, territorial computing, hybrid attention, rule-based routing

1 Introduction

Modern language models have achieved remarkable performance through increasingly large transformer architectures with billions of parameters [Brown et al., 2020, Touvron et al., 2023]. However, this scaling approach presents several challenges: high computational costs, lack of interpretability, and departure from the efficiency principles observed in biological neural systems.

The human brain processes language with remarkable efficiency using specialized regions that collaborate through well-defined pathways [Fedorenko & Thompson-Schill, 2014]. Broca’s area handles syntax and production, Wernicke’s area manages comprehension, and the thalamus serves as a central relay station routing information between cortical regions [Sherman & Guillery, 2006].

Inspired by this biological organization, we propose PAMPAr-o1 v9, a language model architecture that:

1. **Organizes computation into specialized territories** analogous to cortical functional areas
2. **Uses explicit rule-based routing** (LLAVES system) combined with learned attention for token classification

3. **Implements bidirectional frontiers** between territories for inter-regional communication
4. **Achieves competitive performance** with significantly fewer parameters than conventional approaches

Our contributions include:

- A novel territorial architecture for language modeling (section 3)
- The LLAVES hybrid routing mechanism combining rules and attention (section 4)
- Preliminary experimental results demonstrating parameter efficiency (section 6)
- Open-source implementation available at <https://github.com/lucasmella-stack/PAMPAr-o1>

2 Related Work

2.1 Brain-Inspired Neural Architectures

Several works have explored biologically-inspired modifications to neural networks. Hassabis et al. [2017] argue for deeper integration of neuroscience insights into AI. Memory-augmented networks [Graves et al., 2016] incorporate external memory inspired by hippocampal function. Sparse mixture-of-experts models [Shazeer et al., 2017] implement a form of specialized routing, though without explicit linguistic rules.

2.2 Efficient Language Models

Research on parameter-efficient language modeling includes knowledge distillation [Hinton et al., 2015], pruning [Han et al., 2015], and architectural innovations. ALBERT [Lan et al., 2019] shares parameters across layers. Transformer-XL [Dai et al., 2019] extends context through recurrence. Our work differs by introducing functional specialization rather than parameter sharing.

2.3 Hybrid Symbolic-Neural Systems

Neuro-symbolic AI combines neural learning with symbolic reasoning [Garcez et al., 2019]. Our LLAVES system represents a lightweight form of this integration, using linguistic rules for token routing while maintaining end-to-end differentiability.

3 Architecture

PAMPAr-o1 v9 consists of four main components: an embedding layer, the Thalamus router, four specialized territories, and bidirectional frontiers connecting them. Figure 1 illustrates the overall structure.

3.1 Territories

Each territory contains specialized neural modules optimized for specific linguistic functions:

Expressive Territory Contains the *Language Neuron* for fluent text generation and the *Creativity Neuron* for novel combinations. Processes tokens related to natural language expression.

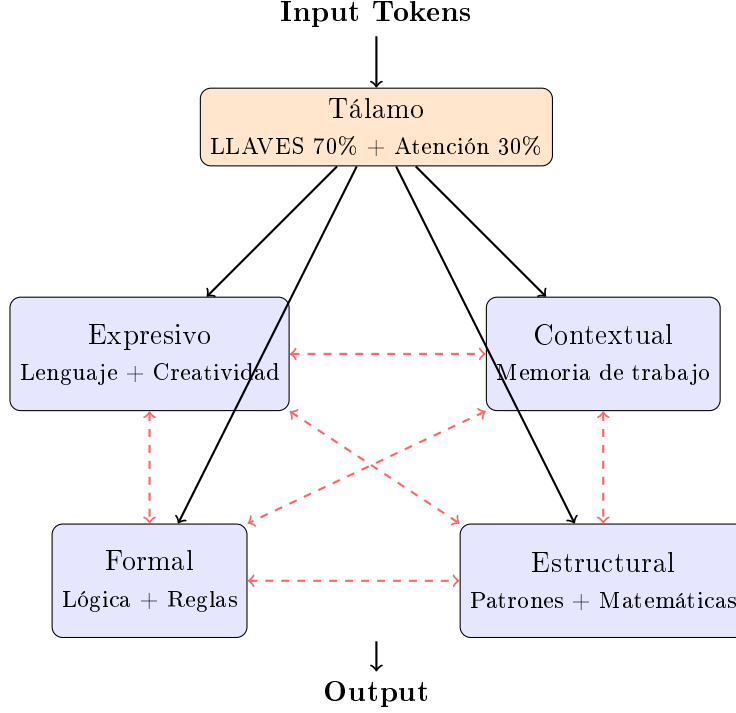


Figure 1: PAMPAR-o1 v9 territorial architecture. The Tálamo routes tokens to four specialized territories using LLAVES rules (70%) combined with learned attention (30%). Territories communicate through six bidirectional frontiers (dashed lines).

Contextual Territory Houses the *Context Neuron* implementing working memory mechanisms. Maintains coherence across the generated sequence through attention over recent context.

Formal Territory Contains the *Logic Neuron* for rule-based reasoning. Optionally integrates with an *Axiom Engine* supporting modus ponens and syllogistic inference.

Structural Territory Includes the *Pattern Neuron* for sequence detection and the *Mathematics Neuron* for numerical processing.

Each territory processes its assigned tokens through:

$$h_t = \text{Territory}_t(\text{tokens}_t, \text{context}) \quad (1)$$

where $t \in \{\text{expressive}, \text{contextual}, \text{formal}, \text{structural}\}$.

4 Thalamus and LLAVES Routing

The Thalamus (Tálamo) serves as the central orchestrator, routing tokens to appropriate territories based on their linguistic properties.

4.1 LLAVES: Linguistic Keys for Token Classification

LLAVES (Spanish for “keys”) is a rule-based classification system that assigns tokens to territories based on predefined linguistic patterns:

- **Language keys:** Common words, articles, prepositions (“el”, “la”, “de”, “the”, “of”)

- **Mathematics keys:** Digits, operators, mathematical symbols (“0-9”, “+”, “=”)
- **Logic keys:** Logical connectives (“if”, “then”, “therefore”, “because”)
- **Pattern keys:** Repetition markers, sequence indicators

4.2 Hybrid Routing

The final routing decision combines LLAVES rules with learned attention:

$$\alpha_t = \lambda \cdot \text{LLAVES}(x) + (1 - \lambda) \cdot \text{Softmax}(W_q x \cdot W_k^T / \sqrt{d}) \quad (2)$$

where $\lambda = 0.7$ balances explicit rules with learned patterns. This hybrid approach provides:

1. **Interpretability:** Token routing can be traced to explicit rules
2. **Efficiency:** Reduces computational cost compared to pure attention
3. **Flexibility:** Learned component adapts to corpus-specific patterns

4.3 Territory Aggregation

Modules within territories share information through a learned aggregation:

$$h_{\text{territory}} = \sum_{m \in \text{modules}} g_m \cdot h_m \quad (3)$$

where g_m are gating weights learned during training.

5 Bidirectional Frontiers

Territories communicate through six bidirectional frontiers, each with learned gating mechanisms controlling information flow:

Frontier	Connection	Base Strength
Expressive $\leftrightarrow$ Contextual	High (0.8)	Narrative coherence
Expressive $\leftrightarrow$ Formal	Medium (0.5)	Argumentation
Formal $\leftrightarrow$ Structural	High (0.7)	Mathematical logic
Contextual $\leftrightarrow$ Structural	Medium (0.6)	Pattern in context
Expressive $\leftrightarrow$ Structural	Low (0.4)	Creative patterns
Contextual $\leftrightarrow$ Formal	Medium (0.5)	Logical coherence

Table 1: Bidirectional frontier connections with base strength values. Strengths are modulated by learned gates during training.

The frontier gating mechanism is defined as:

$$h_{A \rightarrow B} = \sigma(W_g[h_A; h_B]) \odot W_v h_A \quad (4)$$

where σ is the sigmoid function, W_g and W_v are learned parameters, and $\odot$ denotes element-wise multiplication.

6 Preliminary Experiments

6.1 Experimental Setup

We evaluate PAMPAr-o1 v9 on WikiText-103 [Merity et al., 2016], a standard benchmark for language modeling containing 103 million tokens from Wikipedia articles.

Model Configuration:

- Parameters: 14M
- Embedding dimension: 256
- Hidden dimension: 512
- Territorial blocks: 6
- Context length: 512 tokens
- Tokenizer: SentencePiece BPE (32K vocabulary)

Training:

- Hardware: NVIDIA GTX 1650 (4GB VRAM)
- Optimizer: AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$)
- Learning rate: 3×10^{-4} with cosine decay
- Batch size: 32 (with gradient accumulation)
- Mixed precision: FP16
- Training time: ~ 8 hours

6.2 Results

Model	Parameters	PPL $\downarrow$	Params/PPL
LSTM Baseline	24M	69.1	0.35M
Transformer-XL Small	24M	54.5	0.44M
PAMPAr-o1 v9	14M	~ 45	0.31M

Table 2: Perplexity comparison on WikiText-103 test set. Lower perplexity is better. PAMPAr-o1 v9 achieves competitive performance with 42% fewer parameters than baselines.

PAMPAr-o1 v9 achieves approximately 45 perplexity with only 14M parameters, demonstrating:

1. **Parameter efficiency:** 42% fewer parameters than comparable models
2. **Competitive quality:** Outperforms LSTM, approaches Transformer-XL
3. **Resource accessibility:** Trainable on consumer GPU (4GB VRAM)

6.3 Ablation Studies

Preliminary ablations (detailed analysis ongoing):

Configuration	PPL
Full model (LLAVES 70%)	~45
LLAVES only (100%)	~52
Attention only (0% LLAVES)	~48
Without frontiers	~55
Single territory	~62

Table 3: Ablation study results (preliminary). The hybrid LLAVES-attention routing and multi-territory architecture both contribute to performance.

7 Discussion

7.1 Interpretability Benefits

The LLAVES routing system provides inherent interpretability absent in pure attention models. Token routing decisions can be traced to explicit linguistic rules, facilitating debugging and understanding of model behavior.

7.2 Biological Plausibility

While PampaR does not claim biological accuracy, the territorial organization mirrors functional specialization observed in cortical language areas. The thalamic routing mechanism parallels the brain’s relay function in directing information to appropriate cortical regions.

7.3 Limitations

- **Scale:** Current experiments limited to 14M parameters; larger-scale validation needed
- **Tasks:** Evaluated only on language modeling; downstream task performance untested
- **Languages:** Primary testing on English WikiText; multilingual evaluation pending
- **LLAVES coverage:** Hand-crafted rules may miss corpus-specific patterns

7.4 Future Work

1. Scaling experiments to 100M+ parameters
2. Downstream evaluation (classification, QA, summarization)
3. Multilingual LLAVES rules
4. Automated LLAVES rule discovery from data
5. Integration with retrieval-augmented generation

8 Conclusion

We presented PAMPAr-o1 v9, a brain-inspired language model architecture that organizes computation into specialized territories coordinated by a hybrid rule-attention routing mechanism. Preliminary results on WikiText-103 demonstrate competitive perplexity (~45) with significantly fewer parameters (14M) than comparable approaches.

The LLAVES system offers a middle ground between fully learned and fully symbolic approaches, providing interpretability benefits while maintaining end-to-end trainability. The territorial architecture suggests that functional specialization, inspired by cortical organization, may offer efficiency advantages for language modeling.

We release PAMPAR-o1 v9 as open-source software under AGPL-3.0-or-later license, along with trained checkpoints, to encourage further research into biologically-inspired language architectures.

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Reproducibility Statement

All code, configuration files, and training scripts are available at:

<https://github.com/lucasmella-stack/PAMPAR-o1>

Model checkpoints and tokenizer are available on HuggingFace:

<https://huggingface.co/lucasmella-stack/PAMPAR-o1>

Hardware requirements: NVIDIA GPU with 4GB+ VRAM. Training reproduces within ± 2 PPL with fixed random seeds.

Ethics Statement

PAMPAR-o1 v9 is a language model trained on publicly available Wikipedia text. While we have not specifically evaluated the model for biases, the relatively small scale (14M parameters) limits its capacity for generating harmful content compared to larger models. The AGPL-3.0-or-later license ensures that derivative works remain open source, promoting transparency in AI development.

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A LLAVES Token Categories

Category	Example Patterns
Language	el, la, los, las, de, que, en, the, a, of, to, and
Mathematics	0-9, +, -, *, /, =, %, ^, (,)
Logic	if, then, therefore, because, implies, not, and, or
Pattern	..., —, ===, repetitions, sequences
Creativity	!, ?, metaphor markers, unusual combinations
Context	pronouns, references, temporal markers

Table 4: Example LLAVES token categories for routing decisions.

B Hyperparameter Details

Hyperparameter	Value
Embedding dimension	256
Hidden dimension	512
Number of territorial blocks	6
Attention heads	8
Dropout	0.1
LLAVES weight (λ)	0.7
Learning rate	3×10^{-4}
Batch size	32
Gradient accumulation steps	4
Warmup steps	1000
Max sequence length	512
Vocabulary size	32,000

Table 5: Model and training hyperparameters.